

Data Visualization and Hadoop

Unit-6

Unit VI	Data Visualization and Hadoop	(7 Hours)
Introduction to Data Visualization, Types of data visualization, Data Visualization Techniques, Tools used in Data Visualization, Challenges to Big data visualization, Visualizing Big Data, Analytical techniques used in Big data visualization, Hadoop ecosystem, Map Reduce, Pig, Hive,. Data Visualization using Python: Line plot, Scatter plot, Histogram, Density plot, Box- plot.		
#Exemplar/Case Studies	Use IRIS dataset from Scikit and plot 2D views of the dataset	
Mapping of Course Outcomes for Unit VI	CO5, CO6	

Introduction to Data Visualization

- ← Data visualization is the graphical representation of information and data.
- ← It uses visual elements such as charts, graphs, and maps to help viewers understand the significance of data by presenting it in a visual context.
- ← The primary goal of data visualization is to communicate complex data clearly and effectively, allowing users to identify trends, patterns, and relationships within the data.

Data visualization serves several purposes:

- ← **Exploration:** It enables analysts to explore data sets and discover insights that may not be immediately apparent from raw data.
- ← **Communication:** It facilitates the communication of data-driven insights to various stakeholders, including decision-makers, clients, and the general public.
- ← **Analysis:** It supports data analysis by providing visual tools for examining data relationships, distributions, and outliers.
- ← **Presentation:** It enhances presentations and reports by making data more engaging and understandable to the audience.

<https://datavizcatalogue.com/>

- ← **Bar Charts:** Used to compare categorical data or show changes over time.
- ← **Line Charts:** A line chart is a graph that uses **points connected by straight lines** to Display trends over time or relationships between variables.
- ← **Pie Charts:** Show the proportional distribution of categories within a whole.
- ← **Scatter Plots:** Visualize the relationship between two numerical variables.
- ← **Heatmaps:** Represent data values with colors in a matrix format.
- ← **Maps:** Display spatial data, such as geographical distributions or locations.

Types of data visualization

- ← **Bar Charts:** Bar charts are used to compare categories of data. They consist of rectangular bars with lengths proportional to the values they represent. Horizontal bar charts are often used for comparing categories, while vertical bar charts are used for time-series data.
- ← **Line Charts:** Line charts are ideal for showing trends over time or relationships between continuous variables. They connect data points with straight lines, making them useful for visualizing changes and patterns.
- ← **Pie Charts:** Pie charts represent data as slices of a circular "pie," with each slice corresponding to a category's proportion of the whole. They are useful for illustrating the composition of a dataset but can be less effective than other chart types for comparing values.

- ← **Scatter Plots:** Scatter plots display individual data points as dots on a two-dimensional plane, with one variable plotted on each axis. They are used to visualize relationships between two continuous variables, showing patterns such as correlations, clusters, or outliers.
- ← **Histograms:** Histograms display the distribution of a single continuous variable by dividing it into bins and plotting the frequency or density of data points within each bin. They are useful for understanding the shape and spread of data distributions.
- ← **Heatmaps:** Heatmaps represent data values as colors in a matrix, with rows and columns corresponding to two categorical variables. They are often used to visualize patterns or relationships in large datasets, such as correlation matrices or geographical data.

- ← **Bubble Charts:** Bubble charts are similar to scatter plots but use bubble size to represent an additional dimension of data. They are effective for visualizing three variables simultaneously, with the x and y axes representing two variables and bubble size representing a third.
- ← **Treemaps:** Treemaps display hierarchical data as nested rectangles, with each rectangle's size proportional to the data it represents. They are commonly used to visualize hierarchical structures such as file directories, organizational charts, or nested categories.
- ← **Choropleth Maps:** Choropleth maps use color shading or patterns to represent spatial data, such as population density, income levels, or election results, within geographic regions like countries, states, or counties.
- ← **Network Diagrams:** Network diagrams represent relationships between entities as nodes (points) connected by edges (lines). They are used to visualize complex systems, networks, or relationships, such as social networks, organizational structures, or transportation networks.

Data visualization techniques

- ← Data visualization techniques are diverse and versatile, offering various methods to represent data effectively

1. **Charts and Graphs:**

- Bar Charts
- Line Charts
- Pie Charts
- Area Charts
- Scatter Plots
- Bubble Charts
- Histograms
- Box Plots
- Waterfall Charts
- Radar Charts
- Gantt Charts
- Polar Charts
- Funnel Charts
- Error Bars

ERROR DATA

2. Maps:

- Choropleth Maps
- Proportional Symbol Maps
- Dot Density Maps
- Heat Maps
- Cartograms
- Flow Maps
- Network Maps

3. **Hierarchical Visualization:**

- Tree Diagrams
- Sunburst Charts
- Treemaps
- Radial Tree Maps
- Partition Diagrams

4. **Temporal Visualization:**

- Time Series Charts
- Calendar Heatmaps
- Streamgraphs
- Timeline Charts
- Event Sequences

5. **Multidimensional Visualization:**

- Parallel Coordinates
- Radar Charts
- Star Plots
- Scatterplot Matrices

- Scatterplot Matrices

6. **Text Visualization:**

- Word Clouds
- Tree Maps
- Text Scatter Plots
- Sentiment Analysis Visualization

7. **Interactive Visualization:**

- Interactive Dashboards
- Zoomable Interfaces
- Drill-Down Charts
- Hover Effects
- Tooltips

8. **Network Visualization:**

- Node-Link Diagrams
- Arc Diagrams
- Matrix Diagrams
- Force-Directed Graphs

9. **Statistical Visualization:**

- QQ Plots
- Violin Plots
- Probability Density Plots
- Survival Curves
- Pareto Charts

10. **Spatial Visualization:**

- 3D Visualization
- Virtual Reality (VR) Visualization
- Augmented Reality (AR) Visualization

11. **Interactive Storytelling:**

- Story Maps
- Interactive Presentations
- Animated Visualizations

12. **Dynamic Visualization:**

- Animated Charts
- Real-Time Dashboards
- Live Data Feeds

- Live Data Feeds

13. **Machine Learning Visualization:**

- Decision Trees
- Clustering Visualization
- Feature Importance Charts
- Model Interpretability Tools

14. **Geospatial Visualization:**

- Geographical Information Systems (GIS)
- Satellite Imagery Visualization
- Terrain Visualization

15. **Augmented Reality (AR) Visualization:**

- Overlaying Data on Real-World Environments
- Data Visualization through AR Headsets

Data Visualization Tools

- ◀ Tableau
- ◀ Looker
- ◀ Microsoft Power BI
- ◀ Qlik Sense
- ◀ Sisense

Tableau

- ← Tableau is a data visualization tool that can be used by data analysts, scientists, statisticians, etc. to visualize the data and get a clear opinion based on the data analysis.
- ← Tableau is a **powerful data visualization and business intelligence (BI) tool** that helps you **analyze, visualize, and share insights** from your data using **interactive dashboards and charts** — all with **drag-and-drop** simplicity.
- ← Tableau is very famous as it can take in data and produce the required data visualization output in a very short time. And it can do this while providing the highest level of security with a guarantee to handle security issues as soon as they arise or are found by users.
- ← Tableau also allows its users to prepare, clean, and format their data and then create data visualizations to obtain actionable insights that can be shared with other users.
- ← Tableau is available for individual data analysts or at scale for business teams and organizations. It provides a 14-day free trial followed by the paid version.
- ← <https://jason-khu.com/the-tableau-desktop-workspace-interface-guide/>

Looker

- ← Looker is a data visualization tool that can go in-depth into the data and analyze it to obtain useful insights. It provides real-time dashboards of the data for more in-depth analysis so that businesses can make instant decisions based on the data visualizations obtained. It was acquired by **Google Cloud in 2020**.
- ← Entirely web-based, no need to install software. **Real-time insights given by running Queries** directly on your database (no data duplication).
- ← Looker also provides connections with Redshift, Snowflake, and BigQuery, as well as more than 50 SQL-supported dialects so you can connect to multiple databases without any issues.
- ← Looker data visualizations can be shared with anyone using any particular tool. Also, you can export these files in any format immediately. It also provides customer support wherein you can ask any question and it shall be answered.
- ← It uses **LookML (Looker Modeling Language)** A unique feature that allows you to define **data relationships and logic** using code.

Looker

Feature	Looker	Tableau
Data modeling	Uses LookML (code-based)	Mostly drag-and-drop
Deployment	Cloud-first , browser-based	Desktop & Server/Cloud options
Real-time data	Queries data directly in DB	Often imports data for analysis
Ideal for	Data engineers, analysts	Analysts, business users
Integration	Tight with Google Cloud tools	Integrates broadly across platforms

Microsoft Power BI

- ← Microsoft Power BI is a Data Visualization platform focused on creating a data-driven business intelligence culture in all companies today.
- ← To fulfill this, it offers self-service analytics tools that can be used to analyze, aggregate, and share data in a meaningful fashion.
- ← Microsoft Power BI offers hundreds of data visualizations to its customers along with built-in Artificial Intelligence capabilities, which Automatically analyze and explain your data and Excel integration facilities.
- ▶ **Power BI is like the smart assistant for business decision-makers.**
- ▶ **Tableau is the creative canvas for data visualization experts.**

Power BI

- ▶ Power BI is a **powerful business intelligence tool** that helps organizations make **data-driven decisions** by transforming raw data into interactive reports and dashboards.

1. User-Friendly & Easy to Use

- ▶ ☒ Drag-and-drop functionality for creating reports
- ☒ No coding required for basic visualizations
- ☒ Intuitive interface with Excel-like features

4. Advanced Data Modeling & DAX

- ▶ ☒ Supports Excel, SQL Server, Google Analytics, APIs, Azure, etc.
- ☒ Can integrate with cloud and on-premises databases
- ☒ Real-time data streaming for up-to-date insights

Home

Open

Sign in

Options and settings

About

Select a data source or start with a blank report

Blank report

OneLake catalog

Excel workbook

SQL Server

Learn with sample data

Get data from other sources

Recommended

Getting started



Intro—What is Power BI?

Recent

Shared with me

Filter by keyword

Filter

	Name	Location	Opened
	sample.pbix	C:\Users\Vidhya\OneDrive\ドキュメント\sample.pbix	6 days ago

FileHomeInsertModelingViewOptimizeHelp

PasteCutCopyFormat painterClipboard

Get data

Excel workbook catalog

OneLake Server

SQL Enter data

Dataverse

Recent sources

Transform data

Refresh

New visual

Text box

More visuals

New visual calculation

New measure

Quick measure

Sensitivity

Publish

Copilot

Visualizations

Build visual

Filters

Data

Add data to your report

Once loaded, your data will appear in the Data pane.

Import data from Excel

Import data from SQL Server

Paste data into a blank table

Use sample data

Get data from another source

PasteCutCopyFormat painterClipboard

Get dataExcel workbookOneLake catalog

Import data from Excel

Navigator

Search

Display Options

detail.xlsx [1]

details

details

Preview downloaded on Saturday

order_id	Amount	Profit	Quantity	Category	subCategory
1	3040	548	10	Electronics	TV
2	2047	1026	10	Furniture	Chairs
3	4227	2342	10	Clothing	Saree
4	4559	1869	10	Electronics	Printers
5	4879	1762	8	Furniture	Tables
6	2815	1320	7	Clothing	Kurti
7	3525	2233	8	Electronics	TV
8	3882	1173	12	Furniture	Chairs
9	4765	2027	10	Clothing	Saree
10	3281	860	12	Electronics	Printers
11	2755	1880	7	Furniture	Tables
12	2329	693	13	Clothing	Kurti
13	4672	1787	12	Electronics	TV
14	4981	1501	7	Furniture	Chairs
15	4668	1053	7	Clothing	Saree
16	4568	758	13	Electronics	Printers
17	4464	1025	11	Furniture	Tables
18	2368	572	14	Clothing	Kurti
19	4752	1741	9	Electronics	TV
20	2763	1871	9	Furniture	Chairs
21	4925	2263	7	Clothing	Saree
22	4124	2391	14	Electronics	Printers

Load

Transform Data

Cancel

Visualizations

Build visual

Visuals gallery

Filters

Values

Drill through

Data

Search

details

Amount

Category

order_id

Payment mode

Profit

Quantity

subCategory

Build visuals with your data

Select or drag fields from the Data pane onto the report canvas.

Page 1

+

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Manage

Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning

Data Sources Parameters Query Manage Columns Reduce Rows Transform Combine AI Insights

Queries [1]

This preview may be up to 5 days old. Refresh

details

= Table.TransformColumnTypes(#"Promoted Headers",{{"order_id", Int64.Type}, {"Amount", Int64.Type}, {"Profit", Int64.Type}, {"Quantity", Int64.Type}, {"Category", Text.Type}, {"subCategory", Text.Type}, {"Payment mode", Text.Type})

	order_id	Amount	Profit	Quantity	Category	subCategory	Payment mode
1	1	3040	548	10	Electronics	TV	EMI
2	2	2047	1026	10	Furniture	Chairs	credit card
3	3	4227	2342	10	Clothing	Saree	UPI
4	4	4559	1869	10	Electronics	Printers	EMI
5	5	4879	1762	8	Furniture	Tables	credit card
6	6	2815	1320	7	Clothing	Kurti	UPI
7	7	3525	2233	8	Electronics	TV	EMI
8	8	3882	1173	12	Furniture	Chairs	credit card
9	9	4765	2027	10	Clothing	Saree	UPI
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13	13	4672	1787	12	Electronics	TV	EMI
14	14	4981	1501	7	Furniture	Chairs	credit card
15	15	4668	1053	7	Clothing	Saree	UPI
16	16	4568	758	13	Electronics	Printers	EMI
17	17	4464	1025	11	Furniture	Tables	credit card
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21	21	4925	2263	7	Clothing	Saree	UPI
22	22	4124	2391	14	Electronics	Printers	EMI
23	23	3984	1992	13	Furniture	Tables	credit card
24	24	2213	1119	13	Clothing	Kurti	UPI
25	25	4265	1491	12	Electronics	TV	EMI
26	26	2440	1028	8	Furniture	Chairs	credit card

7 COLUMNS, 102 ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name
details

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type

Queries [1] This preview may be up to 5 days old. Refresh

details = Table.TransformColumnTypes(#"Promoted Headers",{"order_id", Int64.Type}, {"Amount", Int64.Type}, {"Profit", Int64.Type}, {"Quantity",

order_id	Amount	Quantity	Category	subCategory	Payment mode
1	1		10	Electronics	TV
2	2		10	Furniture	Chairs
3	3		10	Clothing	Saree
4	4		10	Electronics	Printers
5	5		8	Furniture	Tables
6	6		7	Clothing	Kurti
7	7		8	Electronics	TV
8	8		12	Furniture	Chairs
9	9		10	Clothing	Saree
10	10		12	Electronics	Printers
11	11		7	Furniture	Tables
12	12		13	Clothing	Kurti
13	13		12	Electronics	TV
14	14		7	Furniture	Chairs
15	15		7	Clothing	Saree
16	16		13	Electronics	Printers
17	17		11	Furniture	Tables
18	18		14	Clothing	Kurti
19	19		9	Electronics	TV
20	20		9	Furniture	Chairs
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Query Settings

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- Source
- Navigation
- Promoted Headers
- Changed Type

7 COLUMNS, 102 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON SATURDAY

File Home Insert Modeling View Optimize Help

Clipboard: Paste, Cut, Copy, Format painter

Data: Get data, Excel workbook, OneLake catalog, SQL Server, Enter data, Dataserve, Recent sources

Queries: Transform data, Refresh data

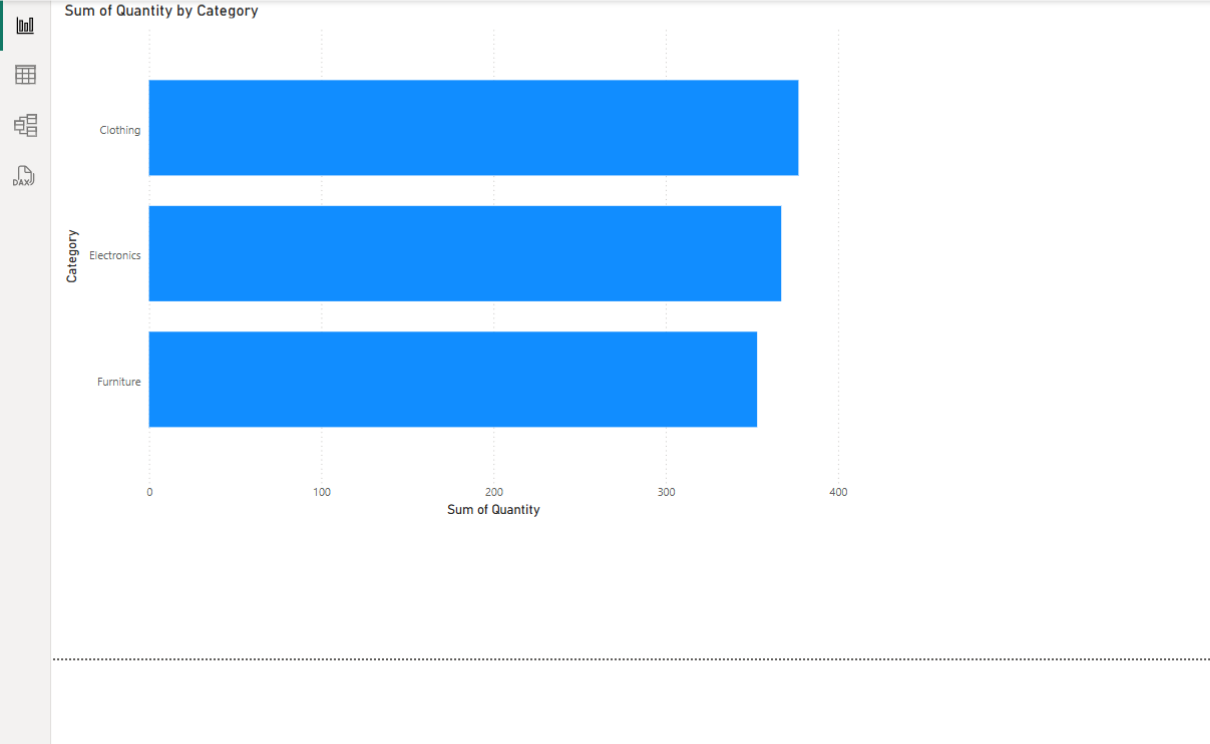
Insert: New visual, Text box, More visuals

Calculations: New visual calculation, New measure, Quick measure

Sensitivity: Sensitivity

Share: Publish

Copilot: Copilot



Filters

Search

Filters on this page: Add data fields here

Filters on all pages: Add data fields here

Visualizations

Build visual

Visuals: [Icons for various chart types]

Py: [Python icon]

Values: Add data fields here

Drill through: Cross-report (Off), Keep all filters (On), Add drill-through fields here

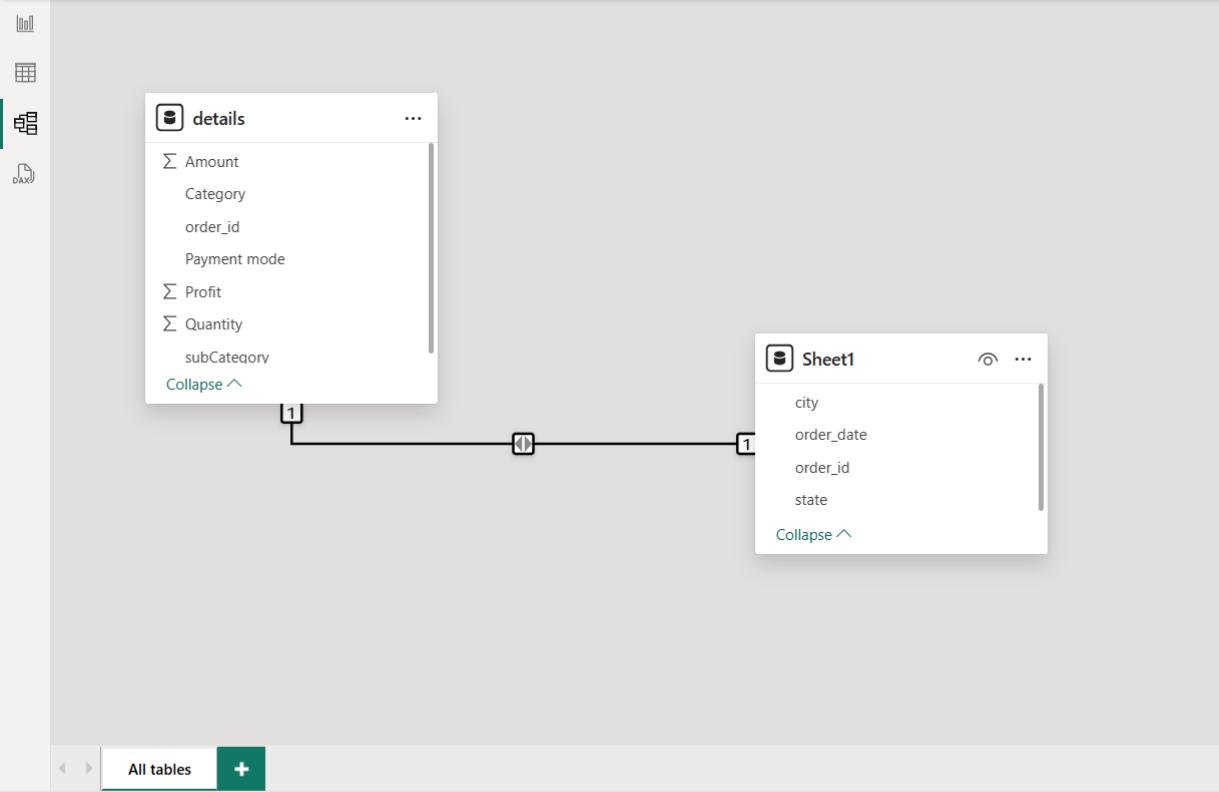
Data

Search

details

- ☐ Amount
- ☐ Category
- ☐ order_id
- ☐ Payment mode
- ☐ Profit
- ☐ Quantity
- ☐ subCategory

Sheet1



Properties

Relationship

Table: Sheet1, Column: order_id

Cardinality: One to one (1:1)

Table: details, Column: order_id

Make this relationship active: Yes

Cross-filter direction: Both

Apply changes, Open relationship editor

Data

TablesModel

Search

details, Sheet1

Feature/Aspect	Power BI	Tableau
Developer	Microsoft	Salesforce
Best For	Business users & Excel pros	Data analysts & visualization experts
Ease of Use	Easier for beginners (especially Excel users)	Slightly steeper learning curve
User Interface	Similar to Microsoft Office	Visual and flexible dashboard interface
Data Handling	Great for smaller to mid-size datasets	Handles very large datasets better
AI & ML Capabilities	Built-in with Microsoft Azure AI	More customizable via Python, R
Integration	Seamless with Excel, Azure, Microsoft tools	Works well with cloud services, databases
Pricing	Cheaper (starts free, Pro = \$10/month)	More expensive (starts ~\$70/user/month)
Deployment Options	Cloud & On-premise	Cloud, On-premise, Tableau Public
Custom Visualization	Fewer customization options	More visual & interactive design flexibility
Community & Support	Strong Microsoft support	Large active data viz community

Qlik Sense

- ◀ **Qlik Sense** is a **self-service data analytics and visualization tool** developed by Qlik. It allows users to **explore data interactively**, build dashboards, and use AI-driven insights.
- ▶ **Associative Data Engine:** Unlike SQL-based tools, Qlik's engine allows you to explore data in any direction – not just through predefined paths.
- ▶ **Smart Search:** Search across the entire dataset like you would on Google.
- ▶ **Augmented Analytics:** Uses AI for pattern detection, insights, and natural language interaction.
- ▶ **Data storytelling:** Present data using slides directly inside the tool.
- ▶ **Strong Governance:** Good for large enterprises with controlled data access.
- ▶ **Multi-cloud & on-premise deployment**

Sisense

- ◀ **Sisense** is a BI platform that helps businesses **embed analytics into applications**. It's known for being **developer-friendly**, supporting **custom analytics solutions**, and handling large, complex datasets.
- ▶ **In-chip technology**: Super-fast processing of large data using your computer's hardware.
- ▶ **Embedded Analytics**: Can embed visualizations into other tools (like apps, websites, CRMs).
- ▶ **API-first design**: Heavy use of REST APIs for customization.
- ▶ **AI & ML Integration**: Works well with Python, R, TensorFlow.
- ▶ **Custom dashboards** with drag-and-drop features.

Sisense

- ▶ Example: Let's say you're building a fitness app. With embedded analytics:
- ▶ You collect user data (steps, calories, workouts),
- ▶ Analyze trends (e.g., progress over weeks),
- ▶ And **embed those personalized reports inside the app dashboard** using Sisense.

Sisense

- ▶ **Best For:**
- ▶ Tech companies or startups that want **analytics within their own products**
- ▶ Developers who want **full control** using APIs
- ▶ Organizations working with **very large datasets**
- ▶ **Qlik Sense:** Great for **business users** who want powerful data exploration without writing code.
- ▶ **Sisense:** Great for **developers** or companies that want to **embed and customize analytics** within their own tools/apps.

Challenges to Big data visualization

- ← *Data visualization is a quick and simple technique to depict complicated ideas graphically for improved comprehension and intuition.*
- ← *It must find diverse relationships and patterns concealed by massive data. We can use traditional graphical representations to organize and represent data.*
- ← *Still, it can be challenging to display Huge amounts of data that is very diverse, uncertain, and semi-structured or unstructured in real-time. Massive parallelization is required to handle and visualize such dynamic data.*

Data Volume (Handling Large Datasets)

Big data consists of **terabytes or petabytes** of information, making traditional visualization tools slow or ineffective.

◆ Example:

- Visualizing **millions of IoT sensor readings** on a single dashboard is difficult.

▶ Possible Solutions:

- ✓ Use **distributed computing tools** (e.g., Apache Spark, Dask).
- ✓ Implement **sampling techniques** to reduce dataset size.

Data Variety (Different Formats & Structures)

Big data comes in **structured (databases)**, **semi-structured (JSON, XML)**, and **unstructured (videos, images, text)** formats, making visualization complex.

- ◆ **Example:**

- Combining **text sentiment analysis (from Twitter)** and **numeric sales data** in a single dashboard.

- ▶ **Possible Solutions:**

- ✓ Use **data integration platforms** (e.g., Apache NiFi, Talend).
- ✓ Convert **unstructured data** into **structured formats** for easier visualization.

Data Quality (Inconsistent and Noisy Data)

Big data often contains **errors, duplicates, missing values, and inconsistencies**, affecting visualization accuracy.

- ◆ **Example:**

- A sales dataset might have **incomplete customer records or incorrect timestamps**.

- ▶ **Possible Solutions:**

- ✓ Perform **data cleaning & preprocessing** (using Pandas, PySpark).
- ✓ Use **machine learning-based anomaly detection** to flag bad data.



← Data Quality

- ← The data quality plays a crucial role in determining its effectiveness. Not all data is created in the same way, and each has a different origin, hence its heterogeneity.
- ← No matter how powerful and comprehensive the big data tools at the organization's disposal are, insufficient or incomplete data can often lead data scientists to conclusions that may not be entirely correct and, therefore, could negatively impact business.



← Accuracy

- ← The most important thing is: How accurate is the data? How much can you trust it? Is there certainty about the collection of relevant data? The values in each field of the database must be correct and accurately represent “real world” values.
- ← **Example:** A registered address must be a real address. Names must be spelled correctly.

← Completeness

- ← The data must contain all the necessary information and be easily understandable by the user.
- ← **Example:** If the first and last name are required in a form, but the middle name is optional, the form can be considered complete even if the middle name is not entered.



← Consistency

- ← The data must be the same throughout the organization and in all systems.
- ← **Example:** The data of a sale registered in the CRM of a company must match the data registered in the accounting dashboard that you manage.



← Format

- ← The data must meet certain standards of type, format, size, etc.
- ← **Example:** All dates must follow the format DD/MM/YY, or whichever format you prefer. Names should only have letters, no numbers or symbols.

Analytical techniques used in Big data visualization

Technique	Use Case	Examples
Descriptive Analytics	Past Data Analysis	Histograms, Heatmaps
Diagnostic Analytics	Cause Analysis	Correlation, Clustering
Predictive Analytics	Forecasting Trends	Regression, Time-Series Models
Prescriptive Analytics	Decision Making	AI Optimization, Recommendations
Real-Time Analytics	Live Data Processing	Streaming, Dashboards

What is Hadoop?

- Hadoop is an open-source software framework that is used for storing and processing large amounts of data in a distributed computing environment.
- It is designed to handle big data and is based on the MapReduce programming model, which allows for the parallel processing of large datasets.

- 
- ▶ It is part of the Apache Software Foundation and is widely used for big data processing..
 - ▶ Its framework is based on Java programming with some native code in C and shell scripts.

Core Components in Hadoop

► Hadoop consists of several core components:

1. Hadoop Distributed File System (HDFS)

- A distributed storage system that stores data across multiple machines.
- Provides high fault tolerance by replicating data blocks.

2. MapReduce

- A programming model for processing large-scale data in parallel.
- Splits tasks into map (filtering and sorting) and reduce (aggregating results) phases.

3. Yet Another Resource Negotiator (YARN)

- Manages and schedules computing resources in a Hadoop cluster.
- Improves efficiency by allowing multiple applications to run simultaneously.

4. Hadoop Common:

A set of utilities and libraries that support Hadoop's core modules.

Features of Hadoop

- ▶ **Scalability** - Can handle petabytes of data by adding more nodes.
- ▶ **Fault Tolerance** - Automatically recovers from hardware failures.
- ▶ **Cost-Effective** - Uses commodity hardware, reducing infrastructure costs.
- ▶ **Flexibility** - Supports various data formats (structured, semi-structured, and unstructured).

Applications of Hadoop

- ▶ **1. Big Data Analytics:** Hadoop enables organizations to analyze massive datasets efficiently.

Example:

- **Facebook** uses Hadoop to process large volumes of user interaction data for insights into user behavior and targeted advertising.

► 2. Data Warehousing

Hadoop can store and manage large-scale data warehouses.

Example:

- **Yahoo!** stores and analyzes petabytes of user search data using Hadoop to improve search algorithms.

► 3. Machine Learning & AI

Hadoop is used to preprocess and store large datasets for machine learning models.

Example:

- **Netflix** utilizes Hadoop to process and analyze user preferences for personalized recommendations

Hadoop Ecosystem

- ▶ The **Hadoop Ecosystem** is a collection of open-source tools and technologies that work together to process, store, and analyze big data efficiently.
- ▶ It extends Hadoop's capabilities beyond basic storage and processing by adding advanced functionalities like real-time analytics, data ingestion, querying, and machine learning.

The Hadoop Ecosystem



Hadoop User Experience (HUE)



Data
Exchange



Sqoop



Zoo Keeper

Coordination

Pig
Scripting



Hive
SQL



Mahout
ML



Oozie
Workflow



APACHE
HBASE

Hbase

Columnar
data
store

Flume



Log
Control



YARN/Map Reduce V2



Hadoop Distributed File System



Hadoop Ecosystem Tools and Frameworks

- ▶ Storage and Management:
- ▶ **HBase** - A NoSQL database for real-time read/write access to large datasets.
- ▶ **Ozone** - A scalable distributed object storage system designed for big data workloads.

► **Data Processing & Computation:**

- **Hive:** A SQL-like data warehouse for querying large datasets in Hadoop.
- **Pig** - A scripting language for large-scale data processing using a simpler syntax than MapReduce
- **Spark** - A fast, in-memory computing framework for big data analytics and machine learning



The Hadoop ecosystem follows a layered approach: **1 Data Storage Layer** → HDFS, HBase

2 Resource Management Layer → YARN

3 Data Processing Layer → MapReduce, Spark, Hive, Pig

4 Workflow & Scheduling Layer → Oozie, Zookeeper

5 Data Ingestion & Streaming Layer → Flume, Kafka, Sqoop

6 Analytics & Machine Learning Layer → Mahout, MLlib



Real-World Use Cases of the Hadoop Ecosystem

- ▶ **Facebook** → Uses Hadoop for storing and analyzing user activity logs
- ▶ **Netflix** → Uses Spark and Hive for real-time recommendation systems.
- ▶ **Amazon** → Uses HBase and Elasticsearch for fast product searches.

Apache Hive

- ▶ Hive is a **data warehousing tool** that provides an SQL-like interface for querying large datasets stored in Hadoop.
- ▶ It converts **SQL queries (HiveQL)** into MapReduce jobs for execution.
- ▶ Suitable for structured and semi-structured data.

Example 1: Analyzing Customer Sales Data

- ▶ **Scenario:** You have a table `sales_data` stored in Hive, and you want to find the total revenue per product (total income a business earns from selling its goods or services).
- ▶ **Hive Query:**

```
SELECT product_id, SUM(price * quantity) AS  
total_revenue  
FROM sales_data  
GROUP BY product_id;
```


Example 2: Finding Active Users in a Social Media Platform

- ▶ You have a table `user_activity` and want to find users who logged in more than 10 times in a week.

```
SELECT user_id, COUNT(*) AS login_count
FROM user_activity
WHERE event_type = 'login'
GROUP BY user_id
HAVING COUNT(*) > 10;
```

Use Case: Used by Facebook & Twitter for engagement analytics.

Apache Pig

- ▶ Pig is a **high-level scripting language** for processing large-scale data.
- ▶ Uses a language called **Pig Latin**, which is simpler than MapReduce.
- ▶ Suitable for both **structured and unstructured data** (logs, XML, JSON).

Introduction to Apache Pig

- One limitation of MapReduce is that the development cycle is very long. Writing the reducer and mapper, compiling packaging the code, submitting the job and retrieving the output is a time-consuming task.
- Apache Pig reduces the time of development using the multi-query approach. Also, Pig is beneficial for programmers who are not from Java background.
- 200 lines of Java code can be written in only 10 lines using the Pig Latin language. Programmers who have SQL knowledge needed less effort to learn Pig Latin.

- Imagine you have a log file containing website visitor data, and you want to count visits per country.

- Script:

```
logs = LOAD 'web_logs.txt' USING PigStorage(',') AS (ip,  
country, timestamp);
```

```
grouped = GROUP logs BY country;
```

```
visit_count = FOREACH grouped GENERATE group,  
COUNT(logs);
```

```
DUMP visit_count;
```

- Output: gives the number of visits per country.

- ▶ script is converted into a **MapReduce job** and executed on Hadoop.
- ▶ The output gives the **number of visits per country**.

When to Use Pig?

When working with semi-structured or unstructured data.

When writing MapReduce code is too complex.

MapReduce

- ▶ MapReduce is a **low-level programming model** for processing large data in parallel.
- ▶ It consists of two functions
 - ▶ **Map** → Filters and processes data.
 - ▶ **Reduce** → Aggregates results.

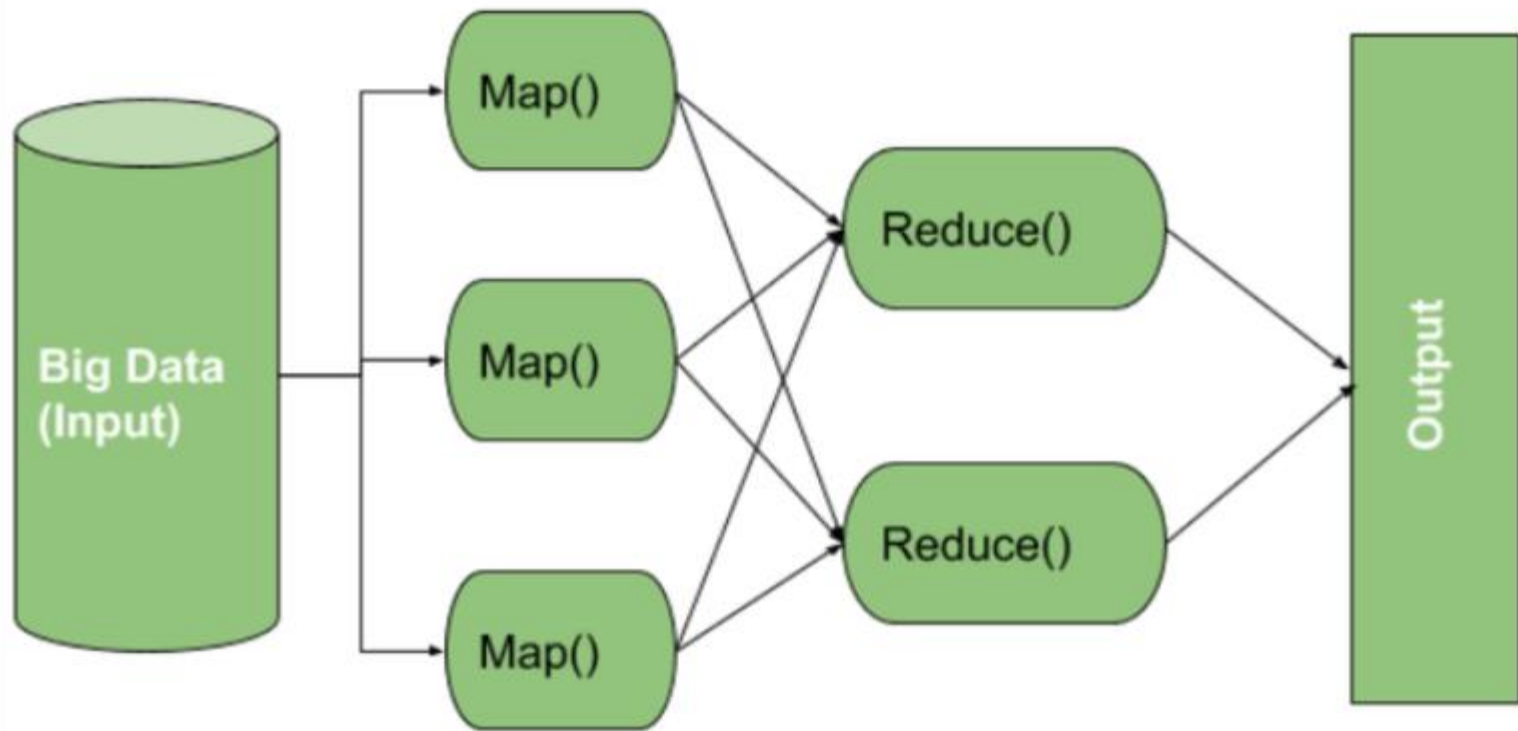
► Map Function (Python Example)::

```
def mapper(key, value):  
    for word in value.split():  
        yield (word, 1)
```

```
def reducer(key, values):  
    yield (key, sum(values))
```

How it Works:

- ▶ The **Mapper** function reads input, breaks text into words, and assigns 1 to each word.
- ▶ The **Reducer** function sums up the occurrences of each word and outputs the final count.
- ▶ **When to Use MapReduce?**
 - ▶ When dealing with large-scale unstructured data.
 - ▶ When high control over computation is needed.



- ▶ Use **Hive** for SQL-like queries on large structured datasets.
- ▶ Use **Pig** for semi-structured/unstructured data when writing SQL is not feasible.
- ▶ Use **MapReduce** when high-level tools (Hive/Pig) are insufficient for complex processing.



**Data Visualization using Python:
Line plot, Scatter plot, Histogram,
Density plot, Box- plot.**

Line Plot

Used for: Showing trends over time.

- ▶ Stock price trends, temperature variations.

```
import matplotlib.pyplot as plt
```

```
# Sample Data
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [10, 15, 7, 20, 18]
```

```
# Line Plot
```

```
plt.plot(x, y, marker='o', linestyle='-', color='b', label='Sales')
```

```
plt.xlabel('Time')
```

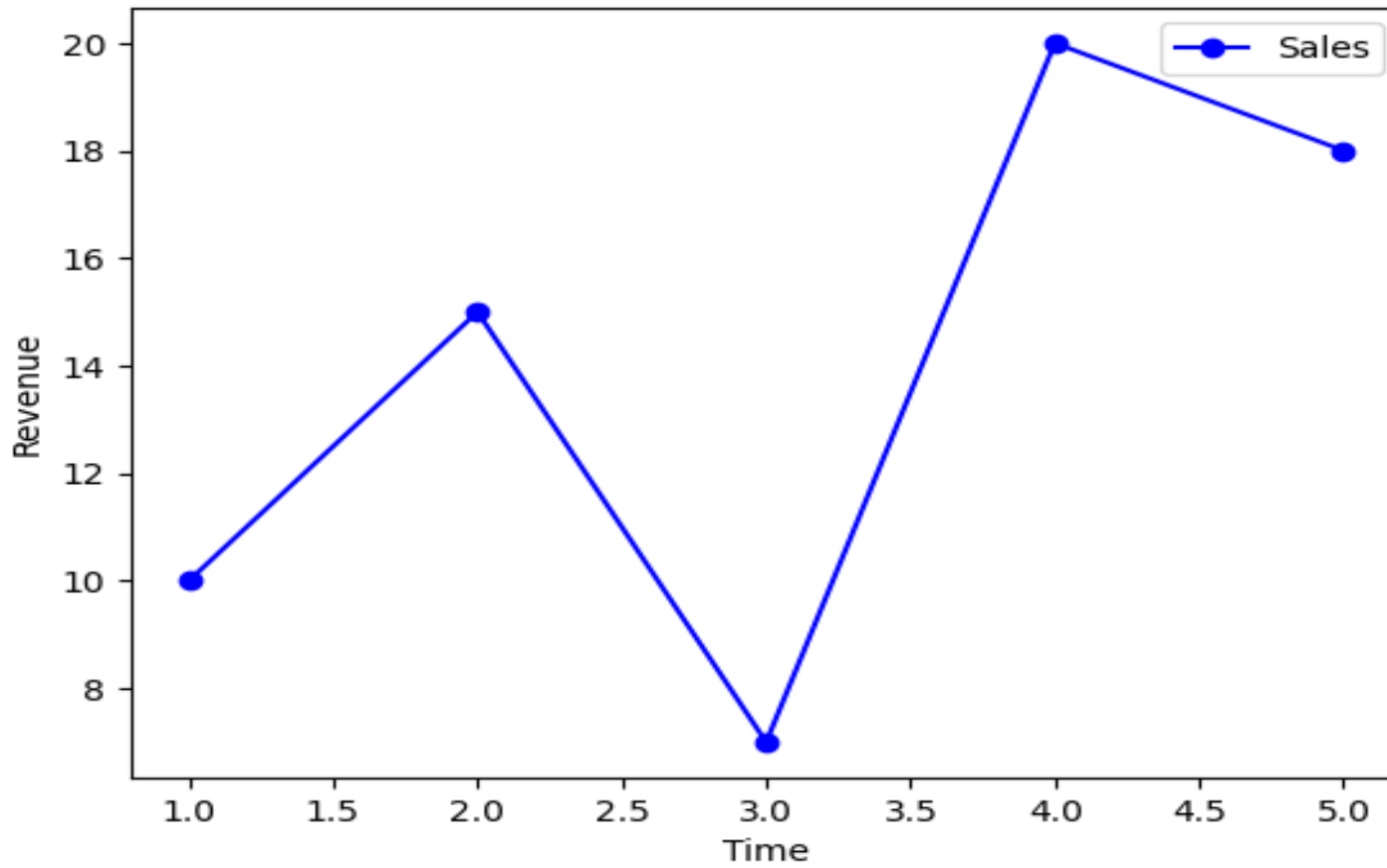
```
plt.ylabel('Revenue')
```

```
plt.title('Sales Over Time')
```

```
plt.legend()
```

```
plt.show()
```

Sales Over Time



Scatter Plot: Showing relationships between two variables.

✓
0s

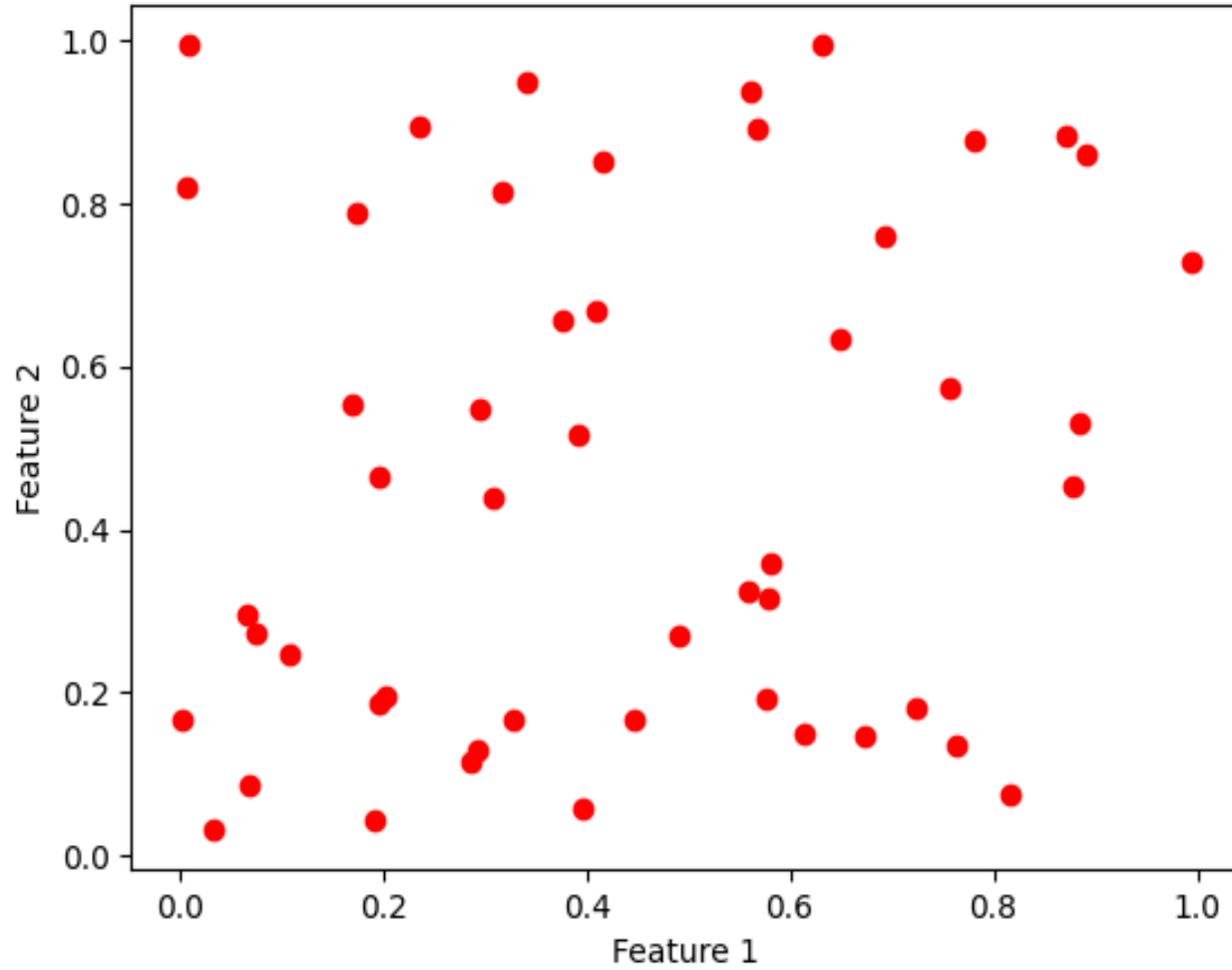


```
import numpy as np

# Random Data
x = np.random.rand(50)
y = np.random.rand(50)

# Scatter Plot
plt.scatter(x, y, color='r', marker='o')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.title('Scatter Plot Example')
plt.show()
```

Scatter Plot Example



Histogram: Displaying the distribution of numerical data.

✓
3s

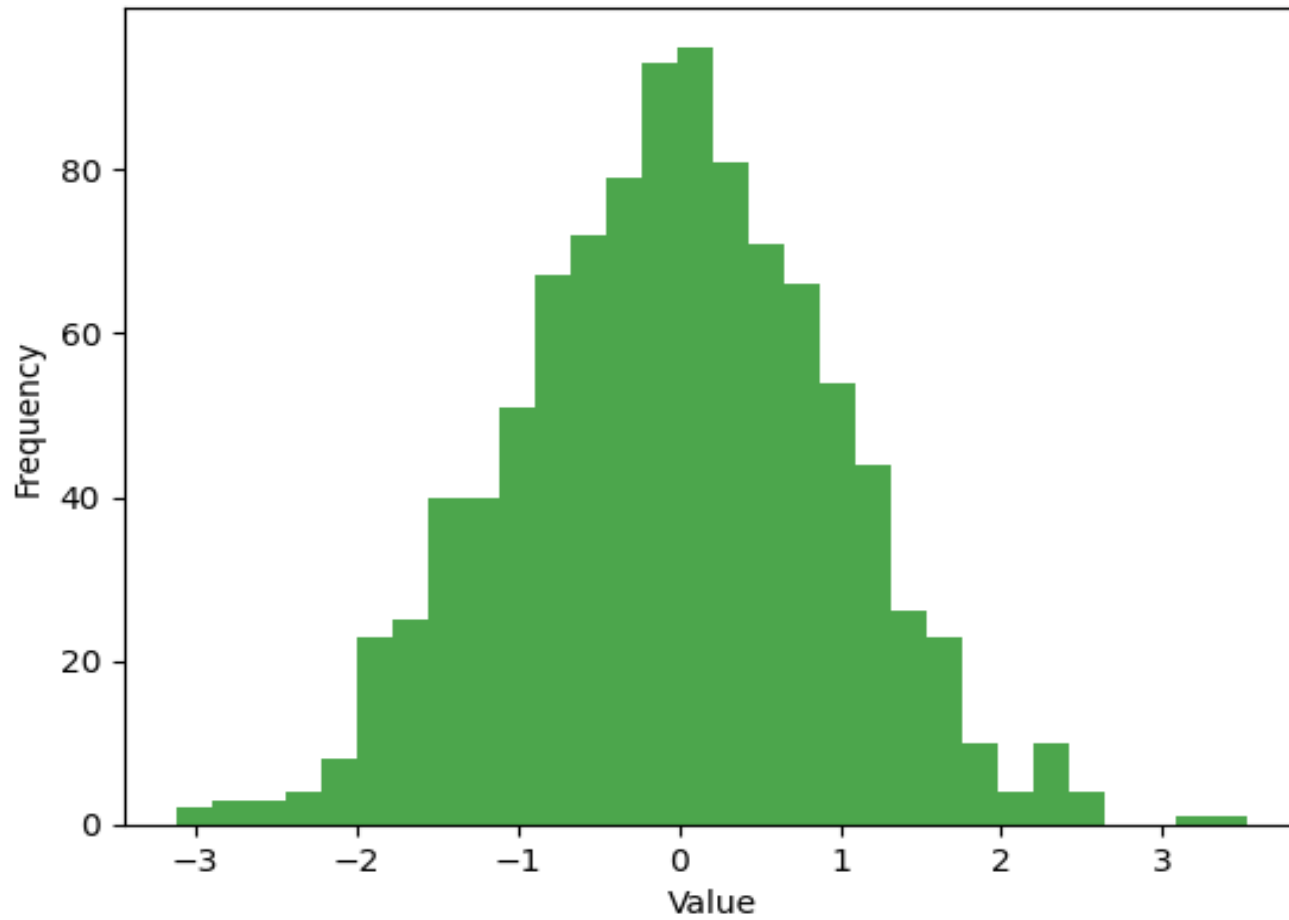


```
import seaborn as sns

# Random Data
data = np.random.randn(1000)

# Histogram
plt.hist(data, bins=30, color='g', alpha=0.7)
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram Example')
plt.show()
```


Histogram Example



Density Plot: Smoothed version of a histogram.

✓
0s



```
# Density Plot
```

```
sns.kdeplot(data, color='purple', fill=True)
```

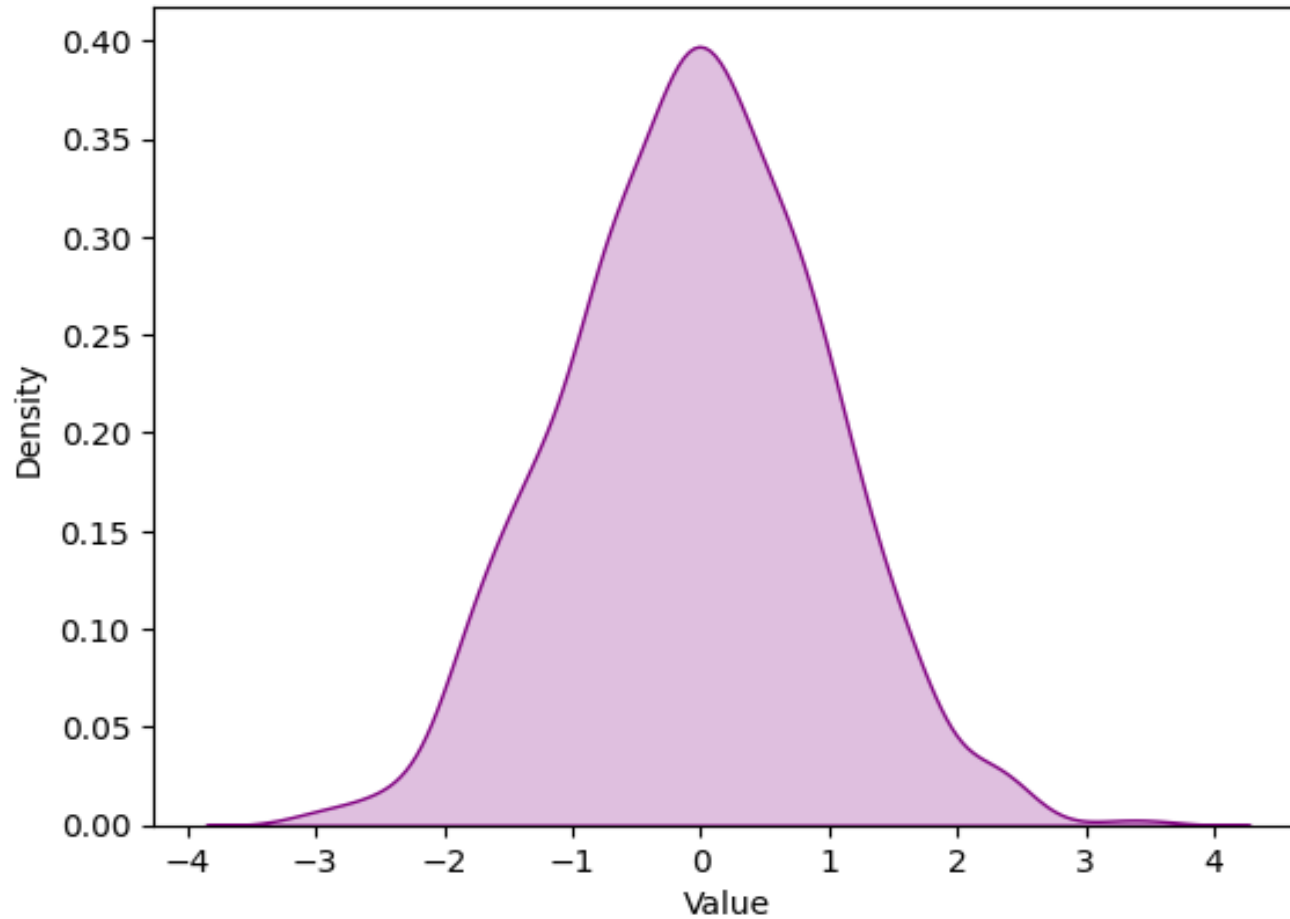
```
plt.xlabel('Value')
```

```
plt.ylabel('Density')
```

```
plt.title('Density Plot Example')
```

```
plt.show()
```

Density Plot Example



Box Plot: Identifying outliers and understanding data spread.

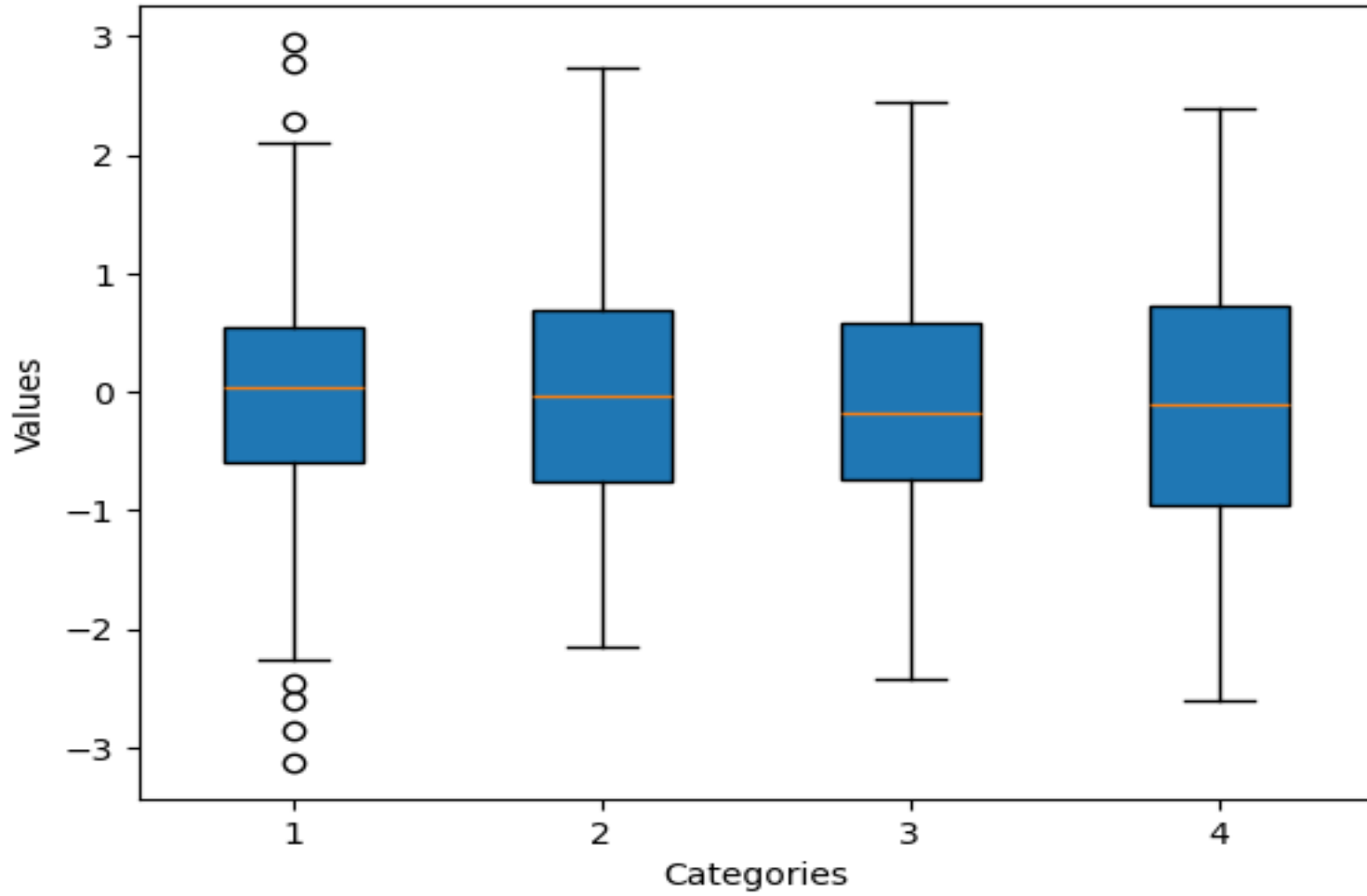
✓
0s



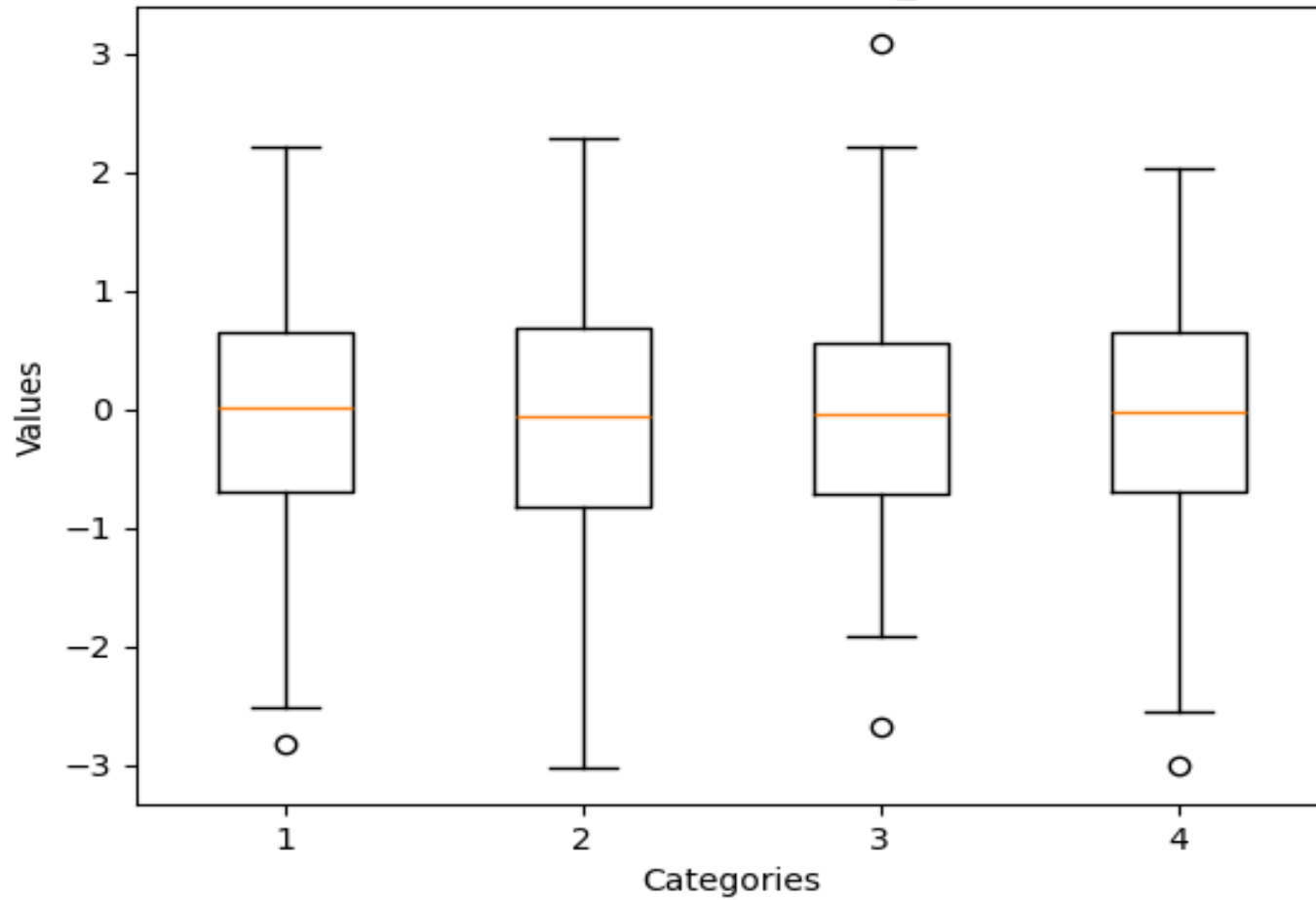
```
# Random Data
data = [np.random.randn(100) for _ in range(4)]

# Box Plot
plt.boxplot(data, patch_artist=True)
plt.xlabel('Categories')
plt.ylabel('Values')
plt.title('Box Plot Example')
plt.show()
```

Box Plot Example



Box Plot Without patch_artist



Questions

- ▶ Write a short note on the following [6]
 - i) Map reduce
 - ii) Pig
 - iii) Hive
- ▶ Describe the Data visualization tool “Tableau”. Explain its applications in brief. [6]
- ▶ With a suitable example explain Histogram and explain its usages. [5]
- ▶ Explain in detail the Hadoop Ecosystem with suitable diagram. [11]
- ▶ With a suitable example explain and draw a Box plot and explain its usages.

Questions

- ▶ What is data Visualization? List and explain any one type of data visualization. [6]
- ▶ With a suitable example explain and draw a Box plot and explain its usage. [6]
- ▶ Discuss various challenges to Big data Visualization. [5]
- ▶ Define data visualization. What are the different methods of data visualization explain in detail. [6]
- ▶ Explain in detail the Hadoop Ecosystem with a suitable diagram. [6]
- ▶ With a suitable example explain the Scatter plot and explain its usage.[5]

Questions

- ▶ What is data visualization? What are the different methods of data visualization explain in detail. [6]
- ▶ Explain in detail the Hadoop Ecosystem with suitable diagram. [11]
- ▶ Describe the Data visualization tool “Tableau”. Explain its applications in brief. [6]
- ▶ With a suitable example explain and draw a Box plot and explain its usages. [6]
- ▶ With a suitable example explain Histogram and explain its usages. [5]